Sixty distinct targets: full-image setup reuse

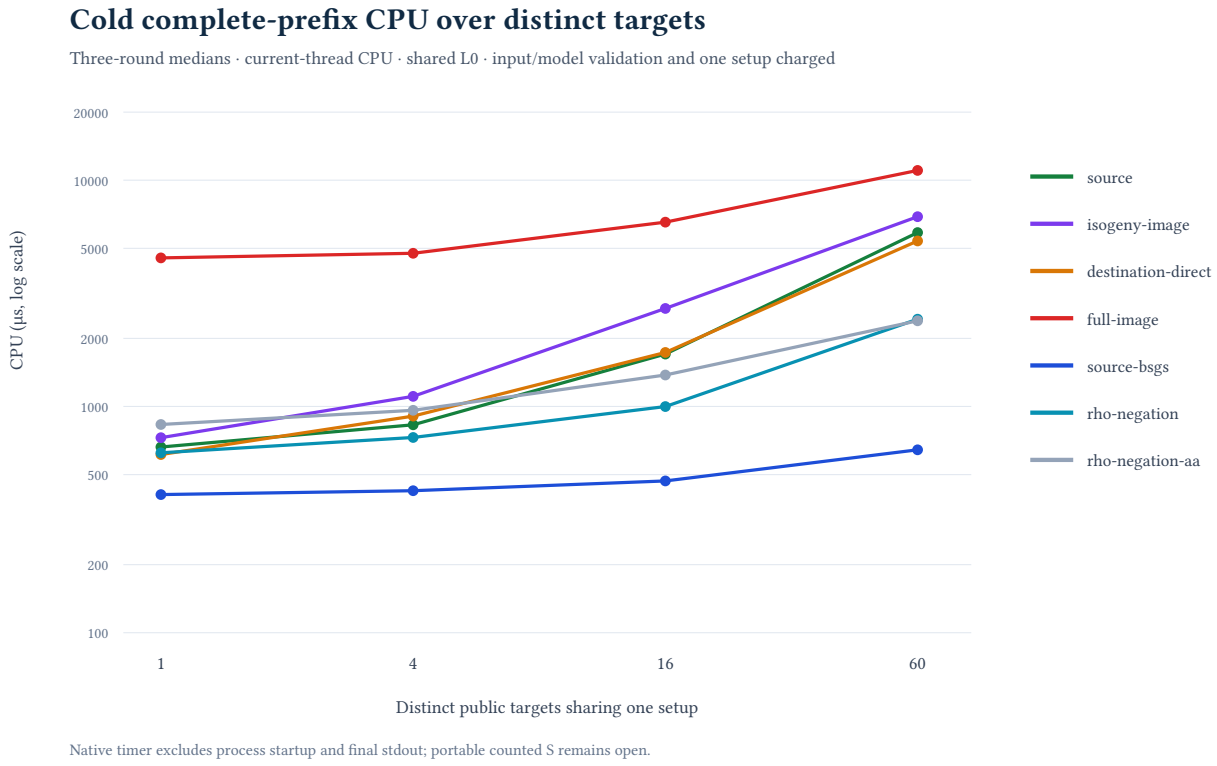
V3 run: 2026-10-15 19:49:52 UTC · **V4 run:** 19:54:56 UTC · **Host:** Darwin arm64, macOS 26.6, shared L0. The explicit degree-three map connects source j=7 to destination j=11 over $F67^2$; the source subgroup has order 61. Public input held all 60 distinct nonidentity targets, ordered by a hash of point coordinates. The solver received no scalar list. Each of seven arms solved all 60 targets in three rounds. Separate Rust arithmetic replays checked **1,260/1,260** scalars and **10,960** relation witnesses in each run.

The V1 scalar-order rule exposed the answers and was corrected before measurement. The V2 shell wrapper then stopped after one arm because zsh reserves `status`; its raw output and stderr are preserved but excluded from comparisons. V3 completed the native-timer panel. V4 repeated the same binary and targets with a child-process CPU, wall and RSS meter; all 21 meter receipts matched raw solver files and replay.

The one table uses **CPU microseconds**. V3 values are cold cumulative current-thread CPU after input/model validation and one setup. V4 values include complete solver startup and final stdout at N=60. Each entry is the median of three rounds.

Arm	V3 N=1	V3 N=4	V3 N=16	V3 N=60	V4 process N=60
Source projected	661.666	829.875	1,704.292	5,872.041	15,199
Isogeny image	727.875	1,108.708	2,711.000	6,900.125	12,735
Destination direct	613.417	907.042	1,732.584	5,389.917	14,021
Full rational image	4,538.750	4,752.500	6,528.750	11,049.667	15,351
Direct source BSGS	407.833	424.500	468.833	643.208	6,148
Direct negation rho	624.125	729.459	999.167	2,431.542	6,908
Rho A/A	832.792	961.625	1,377.625	2,392.959	8,028

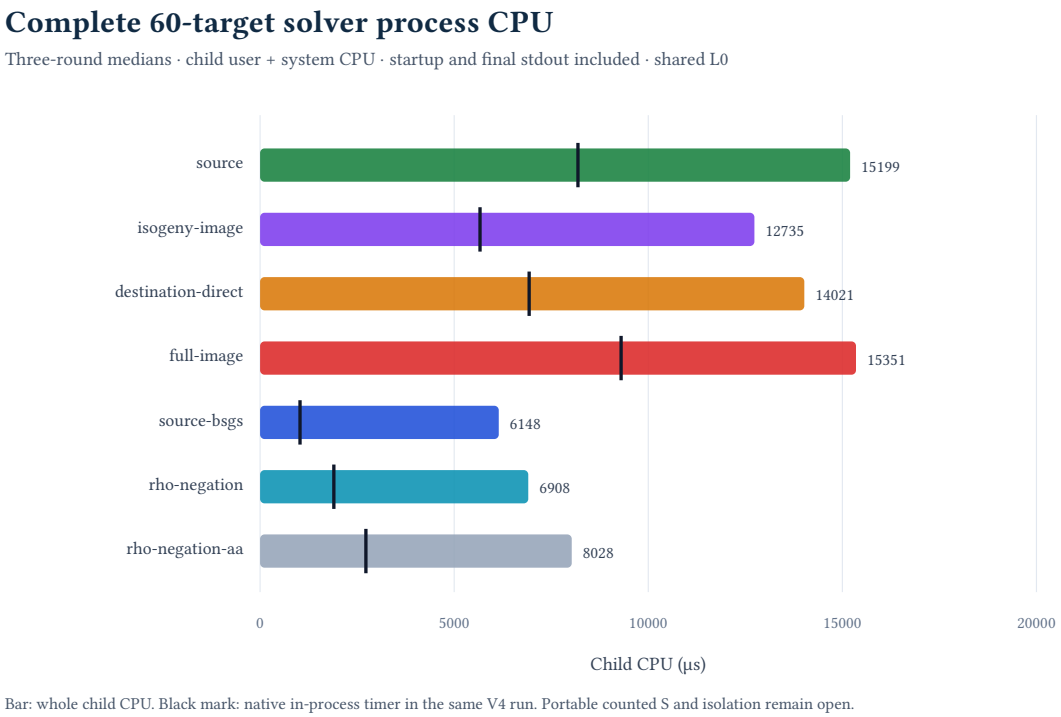
Paired V3 full-image/BSGS CPU ratio medians at N=1,4,16,60 were **11.039, 11.104, 13.926, 18.103**. Against rho they were **7.213, 6.462, 6.534, 5.284**. The N=60 full-image setup median was 4,095.917 µs; its total median was 11,049.667 µs. The per-target total was 184.161 µs at N=60, while BSGS was 10.720 µs per target.



Native timer excludes process startup and final stdout; portable counted S remains open.

Whole-process cost and interpretation

In V4, paired full-image/BSGS child CPU ratios were 2.196, 2.483 and 3.473 (median **2.483**); paired full-image/rho ratios were 2.222, 2.166 and 2.906 (median **2.222**). Full-image median child CPU was 15,351 µs, versus 6,148 µs BSGS and 6,908 µs rho. The child-minus-native-timer medians were 6,330.083, 5,120.250 and 4,788.666 µs, respectively; those differences include startup, output and clock-scope differences. Full-image peak child RSS was at most 9,764,864 bytes; its median wall time was 26.344 ms.



Bar: whole child CPU. Black mark: native in-process timer in the same V4 run. Portable counted S and isolation remain open.

One complete rational-image setup served 60 distinct targets. Full-image still cost more than both direct source controls on this frozen group. The different two-sum support membership did not lower complete solve cost here. V3 prefix values exclude startup and final output; V4 meters the complete N=60 child only. Shared-host L0 timing, one subgroup order, no portable counted S, no isolated wall interval and no second-host performance replay limit this result. The source-linked Markdown report records hashes, raw files, primitive phases, the V2 wrapper failure, custody rejection and the next obligations.